Supplementary Online Content

Graham AL, Cha S, Jacobs MA, et al.. A vaping cessation text message program for adolescent e-cigarette users: a randomized clinical trial. *JAMA*. doi:10.1001/jama.2024.11057

This supplementary material has been provided by the authors to give readers additional information about their work.

eAppendix A. Additional Detail About Study Methods

1. Additional Detail About This is Quitting

This is Quitting provides tips and information about quitting for people across the motivational continuum (users do not have to set a quit date) and across levels of nicotine vaping intensity. The program is individually tailored on several dimensions including participant age, enrollment date, quit date (optional), and the user's primary nicotine device (e.g., Puff Bar). Users receive messages for 1 week preceding their quit date and 8 weeks afterward. One to three messages are sent daily until around Day 30 after their quit date, when message frequency decreases to roughly once every other day. Those who do not set a quit date receive 1 to 3 daily messages for 4 weeks focused on building skills and confidence. Roughly a third of messages are interactive (e.g., true/false, multiple choice, free response questions) and ask about things like confidence to quit, program helpfulness, and vaping behavior (see table below).

This is Quitting is positioned as a nonjudgmental, supportive friend. It is grounded in key constructs from social cognitive theory. For example, to establish and reinforce perceived social norms and social support for quitting, many messages are written by other users (edited by staff where appropriate). These messages reference the author and convey that many other young people are also quitting (e.g., "Ashley says, 'You can do it we are all in this together.' You're not the only one who's thought about quitting."). To support observational learning, messages include quitting strategies from other young people (e.g., "Dalton says, 'Remember that stress can be dealt with in other ways! Try meditating or even writing down what the problem is and then figure out solutions.' You dealt with hard things before you started to vape, and you still can."). To grow behavioral capability, messages give concrete, evidence-based skills and strategies (e.g., "Have your friends supported your quitting? Reply YES or NO." If user responds NO, "Practice—like actually say out loud in front of a mirror at home or in your car—how you'll turn down a JUUL if they offer it to you."). Messages about nicotine replacement therapy describe its use and effectiveness for quitting but that it would need to be prescribed by a healthcare provider. Texting keywords such as COPE, TIPS, FEELS, and STRESS delivers on-demand support around cravings/withdrawal, advice, inspiration, and stress management. Sample text messages are included below as illustrative examples.

Different intervention modalities offer different benefits and value propositions to participants. As a text message intervention, This is Quitting is confidential and relatively anonymous. Participants need not disclose their quitting progress to others, and they can choose to share as much or as little as they want about themselves with the program. This is Quitting is also a private experience, where participants get the benefit of advice from other experienced quitters without being required to interact with anyone else while still being able to interact with the program and use it on-demand for support. Text messaging is free, can be used without internet access, and is universally used by young people. Social media-based interventions, by contrast, meet participants organically "where they are" but may require participants to disclose more about themselves than is comfortable, force social interactions with relative strangers, and require consistent internet or smartphone access that may reinforce disparities.

Message type	**Sample text messages from This is Quitting**
Encourage and support	Michelle L. says "I have always told myself that quitting is impossible, but it's not. Each day is brand new and is another accomplishment." No matter how much you use your vape, it's definitely possible to quit. Experts suggest you make tomorrow a practice day: test out how quitting goes, to see what works and what doesn't. If it doesn't go smoothly, that's ok! You're learning how to do this.
Skill- and self-efficacy building exercises	AnnetteMM says "Gather preparation supplies ahead of time. I bought twizzlers and tootsie pops." Tell me what you're going to do to keep your mind/hands busy instead of a vape next week.
Coping strategies	LizzieG says "Breathe slowly through each craving consciously; knowing you are stronger with every breath." Google "healthy ways to handle stress" if you're needing other ways to deal. Or reply COPE anytime.
Risks of vaping	Pam loves to walk and hike, and her [device] felt like it got her heart racing or she felt nervous. Quitting will help you feel like the best version of you.
Benefits of quitting	Brittany suggests keeping track of how much money you used to spend on your [device] so you can see what you're saving. Makes you feel rich!
COPE	Esirelda says "Reach out for support. My friends & closest family members made a huge impact with supporting me throughout my journey of quitting."
STRESS	Chase says "It's alright take it a few minutes at a time if you are stressed remember it only last a few minutes at a time"
SLIP	Zander says "It's ok to slip up sometimes, the important part thing is that you try again. For most people it can take a couple tries, so don't lose hope!

FEELS	Montana says "Quitting is one of the best decisions I have ever made. It's hard, but it's necessary. Quit for YOU and for the people who love having you around."
TIPS	Jayden says "a helpful tip is to write down a new reason to quit every day, like family, health, or just for yourself."
Vaping behavior (Day 14)	Checking in: Have you cut down how much you vape in the past 2 weeks? Respond w/letter: A=I still vape the same amount, B=I vape less, C=I don't vape at all anymore.
Vaping behavior (Days 30, 60, 90)	How's the quit going? When was the last time you hit a JUUL, or any e-cigarette/vape device, even a puff of someone else's? Respond w/ letter: A= in the past 7 days, B= 8-30 days ago, C= More than 30 days ago.
Confidence	How confident are you feeling about being able to quit? Reply TOTALLY, SOSO or MEH.
Helpfulness	How helpful have these texts been? Reply with a number 1-10, where 10 is the best.

2. Baseline Assessment

All data were collected via self-report. Demographic characteristics, nicotine/e-cigarette dependence, psychosocial characteristics, and other substance use were collected to characterize the study sample and to examine as potential moderators of the treatment-outcome relationship. Demographic questions were drawn from the Centers for Disease Control and Prevention's National Youth Tobacco Survey (NYTS) 2020 and administered as below.

- Grade: What grade are you in?
  - Fixed choice response options (6th, 7th, 8th, 9th, 10th, 11th, 12th, >12th grade (community college or university), Ungraded or other grade, Not a student)
- Gender: Which of the following best describes you?
  - Fixed choice response options (Male, Female, Non-binary, Other, Prefer not to say)
- Race: What race or races do you consider yourself to be? Select one or more.
  - Fixed choice response options (American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian/Other Pacific Islander, White, Other, Prefer not to say)
- Ethnicity: Are you Hispanic, Latino/a, or Spanish origin?
  - Fixed choice response options (Yes, No, Prefer not to say)
- Sexual orientation: Which of the following best describes you?
  - Fixed choice response options (Straight, Gay or lesbian, Bisexual, Other, Prefer not to say)

  Sexual orientation was assessed given its clinical relevance in this study. Sexual minorities have used tobacco at higher rates than their heterosexual counterparts for years. In 2021, lesbian, gay, and bisexual (LGB) young people reported current vaping at higher rates than their heterosexual peers (19.8% vs. 13.2%). Between 2020 and 2021, the difference between LGB and heterosexual young people who currently vape increased by 32%, and those who had ever tried e-cigarettes increased by 26%.[1] Gender- and sexuality-based harassment has been linked to increased e-cigarette use among LGBT high school students.[2] Use of mentholated tobacco products is common in the LGBT community, which increases nicotine addiction levels and difficulty in quitting. The development and testing of tobacco cessation interventions for marginalized communities - including those who identify as LGBTQ+ - is a public health priority and a health equity issue.

eAppendix B: Method Details for the Multiple Imputation Approach

We evaluated the sensitivity of our primary analysis findings to the "missing = vaping" assumption using the multiple imputation (MI) approach of Hedeker et al.[3] which has been further evaluated and refined by others.[4-6] This approach is driven by the observation that in any 2x2 table that stratifies missing outcomes (yes/no) by vaping status at follow-up (yes/no), the odds of vaping among non-respondents can be imputed by first estimating the odds of vaping among respondents and then multiplying them by the odds ratio capturing the association between missingness and vaping (OR.miss). The stronger the odds ratio, the more likely that the missing outcomes represent vaping. Conversely, the weaker the odds ratio, the more likely that the missing data represent abstinence. Two important cases occur when (a) OR.miss=1, i.e., there is no association between missingness and vaping outcome, and (b) OR.miss=+∞, in which case missing outcomes are certain to represent vaping. The first case corresponds to a Missing-at-Random (MAR) scenario, as defined by Little and Rubin[7]. Of note, all OR.miss values other than 1 represent a non-ignorable missingness mechanism (NMAR), whose validity cannot actually be tested from the available data. For that reason, it is common to vary OR.miss over a large range (symmetric around 1 in the logarithmic scale) and to evaluate the sensitivity of the findings to the model assumptions.

To the extent that the odds of vaping among respondents are themselves a random variable subject to estimation uncertainty, multiple imputation is needed to fully propagate this uncertainty to the standard errors of the parameters of interest. In building an imputation model, it is worth noting that missingness is allowed to depend on observed data even under a MAR assumption. It is typical to incorporate baseline predictors of missingness in the imputation model, as they typically have little missing data of their own. Following Hedeker[3] and Smolkowski[4], we chose to enrich our imputation model by allowing the odds of abstinence among observed participants to vary by a dichotomous indicator of nicotine dependence at baseline (a median split on E-cigarette Dependence Scale as the strongest of our nicotine dependence measures). We also evaluated the strength of the (untestable) model assumptions by noting the extent to which it reduces the fraction of missing information (FMI) in the parameter of interest: the odds ratio capturing the association between the text messaging intervention and vaping outcome (OR.vape). Setting all missing outcomes to either vaping (OR.miss=+∞) or abstinence (OR.miss=-∞) reduces the FMI to exactly zero. Therefore, we would expect finite values of OR.miss to reduce the FMI towards zero as they depart from the MAR value (OR.miss=1) on either direction. The FMI is typically unknown but can be estimated as the ratio of the between-imputation variance to the total variance of the parameter of interest, as the between-imputation variance is zero in a scenario with no missing data.[8] To improve the accuracy of the FMI estimate, we used M=10,000 imputations for each value of OR.miss in our sensitivity analyses.

The MI findings are shown in **eTable 3** as we varied OR.miss in a fashion symmetric around 1 in the logarithmic scale in a manner non-differential across study arms. As OR.miss increased from 1/100 to 100, the abstinence rates in the control (P0) and intervention (P1) arms both decreased towards their OR.miss=+∞ estimates, with the difference shrinking monotonically from P1-P0 = 14.16% to 9.92%, indicating a smaller treatment effect as more non-respondents were classified as vapers. Similarly, the use of an odds ratio scale showed treatment effects decreasing from OR=1.83 towards the estimate of OR.miss=+∞ of 1.57. In contrast, using relative risk instead of risk difference as our preferred metric, increases in OR.miss led to an increase in the estimate of the treatment effect from RR=1.26 to their OR.miss=+∞ limit of 1.35. This is because the OR tends to overestimate the RR at high abstinence rates, with the two effect size estimates converging towards each other as abstinence rates drop towards their OR.miss=+∞ values. Despite these differences, all 3 metrics resulted in positive and very highly significant estimates of the treatment effect. For robustness reasons, we chose to highlight the RR finding in the abstract, as it varied over a narrower range than the OR (RR=1.26-1.35 vs. OR=1.57-1.84). Regarding the FMI estimates, we note that they approach zero at both ends of the OR.miss range (1/100, 100), indicating that more extreme values of OR.miss need not be explored. Indeed, estimates of the treatment effect at OR.miss=100 are indistinguishable from the OR.miss=+∞ estimates.

eAppendix C. Inverse Probability of Retention Weighting (IPRW) of Complete Cases

Although widely used in randomized clinical trials for their supposed conservatism, Intention-to-Treat (ITT) analyses that impute all missing outcomes as smoking have been criticized both for being biased and for ignoring the uncertainty associated with the singly imputed values. As an alternative to ITT analyses, Inverse Probability of Retention Weighting (IPRW) has been suggested to correct Complete Case Analyses (CCA) for participants' differential propensity to provide 30-day abstinence data at the 7-month follow-up. Assuming the missingness mechanism is correctly specified, IPRW produces unbiased estimates of the treatment effect under a Missing-At-Random (MAR) assumption.[9] To this effect, we estimated the response rate in each group conditionally on baseline characteristics in **eTable 4**, divided the propensity scores by the overall response rates in each group, and inverted the resulting ratios to create stabilized weights of unit mean under the assumption of no model misspecification. We assessed the success of the IPRW approach in reducing selection bias due to non-response by comparing standardized mean differences (SMD) in baseline characteristics between respondents and non-respondents before and after weighting.[10-12] An SMD of 0.2 pooled standard deviations after weighting indicated that IPRW successfully balanced an individual characteristic.[13] The stabilized weights were then used to estimate survey-weighted logistic regression models for between-arm differences in abstinence outcomes via the *survey* package in R 4.0.2.[14]

As shown in **eTable 5**, 30-day ppa rates at 7-month follow-up under CCA were 55.1% (287/521) in Intervention vs. 38.3% (208/543) among Assessment-only controls (RR=1.44, 95% CI 1.26-1.64, p<.001). CCA estimates of intervention effects were slightly deflated under IPRW (RR=1.42, 95% CI 1.24-1.63; p<0.001). Repeated ppa rates under CCA were 25.3% (131/517) in Intervention and 11.3% (61/538) among Assessment-only controls (RR=2.24, 95% CI 1.70-2.94, p<.001). Again, CCA estimates of intervention effects were slightly deflated under IPWR (RR=2.21, 95% CI 1.67-2.93, p<.001).

To better understand the relationship between ITT and IPRW findings, it is important to understand that IPRW is just a reweighted CCA. Therefore, one can use the CCA-ITT relationship as a starting point. This relationship has been examined in detail by Nelson et al.[15] for all three measures of interest: Risk Difference, Risk Ratio, and Odds Ratio. The authors show that ITT estimates of between-group differences are more conservative (i.e., smaller) than CCA estimates, as long as the difference in the group-specific response rates does not exceed 5%. Given that the 7-month response rates in our study differed by just 4.4% (68.6% in the Intervention group vs. 73.0% in the Assessment-only Control group), it should not be surprising that a CCA would result in a larger estimate of the intervention effect. IPRW suggests that the CCA is too liberal, and pulls back the findings towards the ITT results, with non-respondents more likely to be abstinent than chance alone would allow.

eAppendix D. Method Details for Moderation Analyses

All baseline characteristics described in Table 1 were examined as potential moderators of the treatment effect on the primary outcome (30-day abstinence rates at 7-month follow-up), one moderator at a time. Logistic regression models were fit that included main effects of treatment group, baseline values of the moderator, and treatment by moderator interactions. Logistic regression coefficients (betas) corresponding to the interaction term, their standard errors, and p-values are reported in **eTable 6**.

As the Assessment-only control was used as the reference group in these analyses, betas represent treatment vs. control group differences in the log-odds of abstinence per unit change in the moderator. This interpretation holds for both continuous and binary (0/1) moderators. For multi-category moderators, betas correspond to treatment vs. control group differences in the log-odds of abstinence between the target and reference category of the moderator (e.g., in differences between Male vs. Female). Overall moderation test p-values for multi-category moderators are shown in italics and were used for multiplicity-adjustment purposes using Holm's procedure. P-values for individual contrasts between the target and reference category of multi-category moderators are included for illustrative purposes only and were not used to drive the multiplicity adjustment.

eTable 1. Item Responses to Baseline Measures of Nicotine Dependence and Psychosocial Characteristics

Assessment Instrument	Intervention (n=759)	Assessment Only Control (n=744)
10-item Penn State Electronic Cigarette Dependence Index[a]		
How many times per day do you usually use your electronic cigarette? [No.]	[733]	[727]
0-4 times/day	121 (16.5)	116 (16.0)
5-9 times/day	168 (22.9)	164 (22.6)
10-14 times/day	183 (25.0)	193 (26.5)
15-19 times/day	96 (13.1)	91 (12.5)
20-29 times/day	63 (8.6)	62 (8.5)
30+ times/day	102 (13.9)	101 (13.9)
Vape within 30 minutes of waking?	594 (78.3)	552 (74.2)
Do you sometimes awaken at night to vape?	293 (38.6)	298 (40.1)
If yes, how many nights per week? mean (SD)	3.7 (2.1)	3.4 (2.0)
Do you vape now because it is really hard to quit?	575 (75.8)	542 (72.8)
Do you ever have strong cravings to vape?	646 (85.1)	613 (82.4)
Over the past week, how strong have the urges to vape been?		
No urges/Slight	94 (12.4)	90 (12.1)
Moderate/Strong	487 (64.2)	461 (62.0)
Very Strong/ Extremely Strong	178 (23.5)	193 (25.9)
Hard to keep from vaping in places where you're not supposed to?	533 (70.2)	531 (71.4)
When you tried to stop vaping (or, when you haven't vaped in a while)...		
... did you feel more irritable because you couldn’t vape?	625 (82.3)	631 (84.8)
... did you feel nervous, restless or anxious?	621 (81.8)	599 (80.5)
Hooked on Nicotine Checklist[a]		
Have you ever tried to stop vaping, but couldn’t?	643 (84.7)	627 (84.3)
Do you vape now because it is really hard to quit?	575 (75.8)	542 (72.8)
Have you ever felt like you were addicted to vaping?	681 (89.7)	663 (89.1)
Do you ever have strong cravings to vape?	646 (85.1)	613 (82.4)
Have you ever felt like you really needed to vape?	635 (83.7)	618 (83.1)
Hard to keep from vaping where you're not supposed to?	533 (70.2)	531 (71.4)
When you tried to stop vaping...		
... hard to concentrate because couldn't vape?	538 (70.9)	555 (74.6)
... did you feel a strong need or urge to vape?	655 (86.3)	633 (85.1)
... did you feel more irritable because couldn’t vape?	625 (82.3)	631 (84.8)
... did you feel nervous, restless, anxious because couldn’t vape?	621 (81.8)	599 (80.5)
E-cigarette Dependence Scale[a]		
I find myself reaching for my vape without thinking about it, mean (SD)	2.6 (1.1)	2.5 (1.0)
I drop everything to go out and buy a vape or e-liquids, mean (SD)	1.5 (1.1)	1.5 (1.1)
I vape more before going into situation where vaping not allowed, mean (SD)	2.8 (1.1)	2.8 (1.1)
When not able to vape for a few hours, craving gets intolerable, mean (SD)	2.1 (1.2)	2.1 (1.1)

eTable 1 (continued). Item Responses to Baseline Measures of Nicotine Dependence and Psychosocial Characteristics

Assessment Instrument	Intervention (n=759)	Assessment-only Control (n=744)
E-cigarette Fagerström Test of Cigarette Dependence (e-FTCD)[a]		
How many times/day do you usually use your electronic cigarette? [No.]	[733]	[727]
0-4 times/day	121 (16.5)	116 (16.0)
5-9 times/day	168 (22.9)	164 (22.6)
10-14 times/day	183 (25.0)	193 (26.5)
15-19 times/day	96 (13.1)	91 (12.5)
20-29 times/day	63 (8.6)	62 (8.5)
30+ times/day	102 (13.9)	101 (13.9)
Hard to keep from vaping in places where you're not supposed to?	533 (70.2)	531 (71.4)
When would you hate most to give up vaping?		
In the morning	158 (20.8)	162 (21.8)
During or after meals	88 (11.6)	80 (10.8)
During or after stressful situation	470 (61.9)	451 (60.6)
None of the above	43 (5.7)	51 (6.9)
Vape within 30 minutes of waking?	594 (78.3)	552 (74.2)
Vape more frequently during the first 2 hours of day?	322 (42.4)	302 (40.6)
Vape when so ill that you're in bed most of the day?	402 (53.0)	369 (49.6)
PEdiatric ACEs and Related Life Event Screener (PEARLS)[a] [No.]	[631]	[617]
Adverse Childhood Events		
... lived with a parent/caregiver who went to jail/prison	177 (28.1)	178 (28.8)
... felt unsupported, unloved and/or unprotected	440 (69.7)	410 (66.5)
... lived with a parent/caregiver who had mental health issues	374 (59.3)	375 (60.8)
... been insulted, humiliated, or put down by a parent/caregiver	401 (63.5)	386 (62.6)
... parent/caregiver has/had a drug/alcohol problem	272 (43.1)	261 (42.3)
... lacked appropriate care by any caregiver	182 (28.8)	187 (30.3)
... seen or heard a parent/caregiver being screamed at...	300 (47.5)	269 (43.6)
... been pushed, grabbed, slapped...	217 (34.4)	203 (32.9)
... experienced sexual abuse	237 (37.6)	218 (35.3)
... significant changes in relationship status of caregiver(s)	257 (40.7)	235 (38.1)
Social Determinants of Health		
... seen, heard, or been a victim of violence...	344 (54.5)	347 (56.2)
... experienced discrimination	327 (51.8)	330 (53.5)
... had problems with housing	167 (26.5)	158 (25.6)
... worried that you did not have enough food to eat...	148 (23.5)	147 (23.8)
... been separated from your parent or caregiver...	39 (6.2)	53 (8.6)
... lived with a parent/caregiver with serious physical illness...	83 (13.2)	72 (11.7)
... lived with a parent or caregiver who died	58 (9.2)	53 (8.6)
... been detained, arrested or incarcerated	82 (13.0)	72 (11.7)
... experienced verbal/physical abuse/threats from romantic partner	218 (34.5)	185 (30.0)
Global Appraisal of Individual Needs - Short Screener (GAIN-SS)[a]		
Internalizing disorders, past year		
Significant problems feeling very trapped, lonely, sad...	683 (90.0)	672 (90.3)
Significant problems sleeping	650 (85.6)	656 (88.2)
Significant problems feeling very anxious, nervous, tense...	664 (87.5)	649 (87.2)
Significant problems when something reminded you of the past	662 (87.2)	657 (88.3)
Substance disorders, past year		
Used alcohol or drugs weekly	643 (84.7)	630 (84.7)
Spent a lot of time getting, using, or feeling effects of alcohol or drugs	582 (76.7)	557 (74.9)
Kept using alcohol or drugs even though causing problems	324 (42.7)	329 (44.2)
Use of alcohol or drugs caused problems at work, school, home...	388 (51.1)	392 (52.7)
Experienced withdrawal problems from alcohol or drugs	296 (39.0)	314 (42.2)

[a]All entries are No. (%) of yes/positive responses unless otherwise noted; [when N differs from column No. due to missing values]

eTable 2. Comparison of Baseline Characteristics Between Assessment-only Control and Waitlist Control Groups

	Assessment-only Control	Waitlist Control	SMD
	N=744	N=178	
Demographic characteristics[a]			
Age, mean (SD)	16.4 (0.8)	16.4 (0.8)	0.05
Grade level [No.]	[743]	[177]	0.17
6th - 8th	7 (0.9)	0 (0.0)	
9th	29 (3.9)	7 (4.0)	
10th	129 (17.4)	30 (16.9)	
11th	253 (34.1)	62 (35.0)	
12th	277 (37.3)	70 (39.5)	
>12th Grade	20 (2.7)	4 (2.3)	
Ungraded or Other Grade/Not a student	28 (3.8)	4 (2.3)	
Gender [No.]	[742]		0.08
Female	369 (49.7)	93 (52.2)	
Male	314 (42.3)	69 (38.8)	
Non-binary or Other	59 (8.0)	16 (9.0)	
Sexuality [No.]	[734]	[175]	0.13
LGBQ+	311 (42.4)	85 (48.6)	
Heterosexual	423 (57.6)	90 (51.4)	
Race [No.]	[737]	[171]	0.18
American Indian/Alaskan Native	7 (0.9)	3 (1.8)	
Asian	20 (2.7)	4 (2.3)	
Black	76 (10.3)	15 (8.8)	
Native Hawaiian or Other Pacific Islander	2 (0.3)	1 (0.6)	
White	461 (62.6)	117 (68.4)	
Multiracial	136 (18.5)	26 (15.2)	
Other	35 (4.7)	5 (2.9)	
Hispanic ethnicity [No.]	[735]	[174]	0.08
Yes	117 (15.9)	33 (19.0)	
No	618 (84.1)	141 (81.0)	
Vaping-related items[a]			
Days per month vaping, median (IQR)	30.0 (26.0-30.0)	30.0 (25.0- 30.0)	0.04
Motivation to quit vaping, median (IQR)[b]	4.0 (4.0-5.0)	4.0 (3.0 5.0)	0.07
Confidence to quit vaping, median (IQR)[b]	3.0 (3.0-4.0)	3.0 (3.0-4.0)	0.04
Past year attempts to quit vaping			0.11
None	100 (13.4)	26 (14.6)	
1-2 times	249 (33.5)	63 (35.4)	
3-5 times	302 (40.6)	63 (35.4)	
6 or more times	93 (12.5)	26 (14.6)	
Concern about health consequences of vaping, median (IQR)[b]	3.0 (3.0-4.0)	3.0 (3.0-4.0)	0.02

eTable 2 (continued). Comparison of Baseline Characteristics Between Assessment-only Control and Waitlist Control Groups

	Assessment-only Control	Waitlist Control	SMD
	N=744	N=178	
Nicotine/e-cigarette dependence[a]			
10-item Penn State E-Cigarette Dependence Index, mean (SD)[c] [No.]	11.7 (4.3) [727]	12.0 (4.4) [171]	0.06
E-cigarette Dependence Scale, mean (SD)[d]	8.9 (3.3)	9.2 (3.5)	0.07
E-cigarette Fagerström Test of Cigarette Dependence, mean (SD)[e] [No.]	4.9 (2.3) [727]	4.9 (2.5) [171]	0.01
Hooked on Nicotine Checklist, mean (SD)[f]	8.1 (2.2)	8.2 (2.2)	0.07
Vape within 30 minutes after waking	552 (74.2)	142 (80.0)	0.13
Perceived addiction to vaping			0.04
Very addicted	291 (39.1)	71 (39.9)	
Somewhat addicted	400 (53.8)	96 (53.9)	
Not at all addicted	16 (2.2)	3 (1.7)	
I don't know	37 (5.0)	8 (4.5)	
Psychosocial characteristics[a]			
Global Assessment of Individual Needs - Short Screener, median (IQR)			
Internalizing Disorders[g]	4.0 (3.0-4.0)	4.0 (3.0- 4.0)	0.07
Substance Disorder Problems[h]	3.0 (2.0-5.0)	3.0 (2.0-4.0)	0.05
Loneliness Scale, median (IQR)[i]	8.0 (6.0-10.0)	8.5 (6.0-11.0)	0.10
Pediatric ACEs and Related Life Event Screener (PEARLS)[j] [No.]	[617]	[109]	0.22
High risk	443 (71.8)	87 (79.8)	
Intermediate risk	143 (23.2)	16 (14.7)	
Low risk	31 (5.0)	6 (5.5)	
Other substance use[b]			
Past 30-day use cigarettes	257 (34.5)	49 (27.5)	0.15
Past 30-day use large cigars, little cigars, cigarillos	104 (14.0)	18 (10.1)	0.12
Past 30-day use nicotine pouches	86 (11.6)	17 (9.6)	0.07
Past 30-day use marijuana/cannabis	549 (73.8)	119 (66.9)	0.15

[a] All entries are No. (%) unless otherwise noted; [when N differs from column No. due to missing values]
[b] Range 1-5 (1=not at all, 5=very much)
[c] 10-item Penn State Electronic Cigarette Dependence Index: range 0-20 (0–3=not dependent, 4–8=low dependence, 9–12=medium dependence, 13+=high dependence)
[d] E-cigarette Dependence Scale: range 0-16 (higher score indicates greater dependence)
[e] E-cigarette Fagerström Test of Cigarette Dependence: range 0-10 (0-2=low dependence, 3-4=low to moderate dependence, 5-7=moderate dependence , 8+=high dependence)
[f] 10-item Hooked on Nicotine Checklist: range 0-10 (score greater than 0 shows a loss of some degree of independence over vaping)
[g] Number of internalizing disorder problems (of depression, sleep, anxiety, trauma) experienced in the past 12 months (range 0-4: 0=low severity, 1-2=moderate severity, 3+=high severity)
[h] Number of substance disorder problems experienced in the past 12 months (range 0-5: 0=low severity, 1-2=moderate severity, 3+=high severity)
[i] Roberts UCLA Loneliness Scale: range 0-12 (higher score indicates greater loneliness)
[j] This measure was added to the baseline following study launch (Supplement 1). Missing data are due to timing of administration

eTable 3. Sensitivity of Intervention Effects to Missing Data Assumptions

OR.miss	FMI	P1	P0	Diff.vape	RR.vape	OR.vape	P-val
-∞	0	69.17	54.97	14.20	1.26	1.84	
1/100	2	68.71	54.55	14.16	1.26	1.83	.0001
1/20	6	67.43	53.49	13.94	1.26	1.80	<.0001
1/10	11	65.86	52.15	13.71	1.26	1.77	<.0001
1/5	16	63.14	49.75	13.39	1.27	1.73	<.0001
1/3	19	60.33	47.37	12.96	1.27	1.69	<.0001
1/2	22	57.65	45.06	12.59	1.28	1.66	<.0001
2/3	23	55.48	43.18	12.30	1.28	1.64	<.0001
4/5	23	53.99	41.86	12.13	1.29	1.63	<.0001
1	23	52.26	40.48	11.79	1.29	1.61	.0001
5/4	23	50.59	39.02	11.57	1.30	1.60	.0001
3/2	22	49.39	37.89	11.51	1.30	1.60	.0001
2	20	47.54	36.31	11.24	1.31	1.59	.0001
3	18	45.10	34.21	10.89	1.32	1.58	.0001
5	14	42.46	31.97	10.49	1.33	1.57	.0001
10	9	40.30	30.07	10.23	1.34	1.57	.0001
20	5	39.16	29.08	10.08	1.35	1.57	.0001
100	1	37.98	28.06	9.92	1.35	1.57	.0001
+∞	0	37.98	28.06	9.92	1.35	1.57	.0001

OR.miss = Odds Ratio (OR) capturing the association of vaping and survey non-response
FMI = Fraction of Missing Information in multiple imputation estimate of log(OR.vape)
P1 = abstinence rate (%) in intervention group
P0 = abstinence rate (%) in assessment-only control group
Diff.vape = P1-P0 = difference in abstinence rates between intervention and control arms
RR.vape = P1/P0 = abstinence Rate Ratio (RR) between intervention and control arms
OR.vape = [P1/(1-P1)]/(P0/(1-P0)] = OR capturing the association of abstinence and intervention group

eTable 4. Comparison of Baseline Characteristics Between 7-Month Non-Responders and Responders

	Non-Responder n=439	Responder n=1,064	SMD	P-nom	P-adj
Demographic characteristics[a]					
Age, mean (SD)	16.4 (0.9)	16.4 (0.8)	0.01	.882	1.000
Grade level [No.]	[436]		0.14	.357	1.000
6th - 8th	7 (1.6)	11 (1.0)			
9th	22 (5.0)	35 (3.3)			
10th	74 (17.0)	175 (16.4)			
11th	139 (31.9)	377 (35.4)			
12th	167 (38.3)	411 (38.6)			
>12th Grade	14 (3.2)	21 (2.0)			
Ungraded or Other Grade/Not a student	13 (3.0)	34 (3.2)			
Gender [No.]	[436]	[1,057]	0.22	.001	.025
Female	254 (58.3)	501 (47.4)			
Male	153 (35.1)	475 (44.9)			
Non-binary or Other	29 (6.7)	81 (7.7)			
Sexuality [No.]	[432]	[1,046]	0.13	.028	.364
LGBQ+	203 (47.0)	425 (40.6)			
Heterosexual	229 (53.0)	621 (59.4)			
Race [No.]	[432]	[1,053]	0.28	.002	.034
American Indian/Alaskan Native	5 (1.2)	13 (1.2)			
Asian	7 (1.6)	29 (2.8)			
Black	32 (7.4)	120 (11.4)			
Native Hawaiian or Other Pacific Islander	0 (0.0)	5 (0.5)			
White	301 (69.7)	629 (59.7)			
Multiracial	62 (14.4)	213 (20.2)			
Other	25 (5.8)	44 (4.2)			
Hispanic ethnicity [No.]	[434]	[1,051]	0.05	.433	1.000
Yes	76 (17.5)	165 (15.7)			
No	358 (82.5)	886 (84.3)			
Vaping-related items[a]					
Days per month vaping, median (IQR)	30.0 (27.0-30.0)	29.0 (26.0-30.0)	0.04	<.001	<.025
Motivation to quit vaping, median (IQR)[b]	4.1 (0.8)	4.1 (0.8)	0.08	.125	.960
Confidence to quit vaping, median (IQR)[b]	3.2 (1.1)	3.5 (1.1)	0.25	<.001	<.025
Past year attempt to quit vaping			0.17	.028	.364
None	63 (14.4)	128 (12.0)			
1-2 times	163 (37.1)	346 (32.5)			
3-5 times	150 (34.2)	451 (42.4)			
6 or more times	63 (14.4)	139 (13.1)			
Concern about health consequences of vaping, median (IQR)[b]	4.0 (3.0-5.0)	3.0 (3.0-4.0)	0.08	.120	.960

eTable 4 (continued). Comparison of Baseline Characteristics[a] Between 7-Month Non-Responders and Responders

	Non-Responder n=439	Responder n=1,064	SMD	P-nom	P-adj
Nicotine/e-cigarette dependence[a]					
10-item Penn State E-Cig Dependence Index, mean (SD)[c] [No.]	12.5 (4.3) [428]	11.5 (4.2) [1,025]	0.25	<.001	<.025
E-cigarette Dependence Scale, mean (SD)[d]	9.3 (3.6)	8.8 (3.4)	0.16	.003	.045
e-FTCD, mean (SD)[e] [No.]	5.2 (2.3) [428]	4.8 (2.3) [1,025]	0.19	<.001	<.025
Hooked on Nicotine Checklist, mean (SD)[f]	8.3 (2.1)	8.0 (2.3)	0.13	.053	.530
Vape within 30 minutes after waking	363 (82.7)	783 (73.6)	0.22	<.001	<.025
Perceived addiction to vaping			0.22	.001	.025
Very addicted	212 (48.3)	397 (37.3)			
Somewhat addicted	204 (46.5)	594 (55.8)			
Not at all addicted	8 (1.8)	25 (2.3)			
I don't know	15 (3.4)	48 (4.5)			
Psychosocial characteristics[a]					
GAIN-SS, median (IQR)					
Internalizing Disorders[g]	4.0 (4.0-4.0)	4.0 (3.0-4.0)	0.15	.002	.034
Substance Disorder Problems[h]	3.0 (2.0-5.0)	3.0 (2.0-4.0)	0.05	.236	1.000
Loneliness Scale, median (IQR)[i]	8.0 (5.0-11.0)	8.0 (6.0-10.0)	0.04	.082	.738
Pediatric ACEs and Related Life Event Screener[j] [No.]	[362]	[886]	0.19	.010	.140
Low risk	22 (6.1)	42 (4.7)			
Intermediate risk	58 (16.0)	208 (23.5)			
High risk	282 (77.9)	636 (71.8)			
Other substance use[a]					
Past 30-day use cigarettes	129 (29.4)	370 (34.8)	0.12	.047	.517
Past 30-day use large cigars, little cigars, cigarillos	39 (8.9)	159 (14.9)	0.19	.001	.025
Past 30-day use nicotine pouches	41 (9.3)	119 (11.2)	0.06	.313	1.000
Past 30-day use marijuana/cannabis	329 (74.9)	797 (74.9)	<0.01	1.000	1.000

[a] All entries are No. (%) unless otherwise noted; [when N differs from column No. due to missing values]
[b] Range 1-5 (1=not at all, 5=very much)
[c] 10-item Penn State Electronic Cigarette Dependence Index: range 0-20 (0–3=not dependent, 4–8=low dependence, 9–12=medium dependence, 13+=high dependence)
[d] E-cigarette Dependence Scale: range 0-16 (higher score indicates greater dependence)
[e] e-FTCD (E-cigarette Fagerström Test of Cigarette Dependence): range 0-10 (0-2=low dependence, 3-4=low to moderate dependence, 5-7=moderate dependence , 8+=high dependence)
[f] 10-item Hooked on Nicotine Checklist: range 0-10 (score greater than 0 shows a loss of some degree of independence over vaping)
[g] Number of internalizing disorder problems (of depression, sleep, anxiety, trauma) experienced in the past 12 months (range 0-4: 0=low severity, 1-2=moderate severity, 3+=high severity)
[h] Number of substance disorder problems experienced in the past 12 months (range 0-5: 0=low severity, 1-2=moderate severity, 3+=high severity)
[i] Roberts UCLA Loneliness Scale: range 0-12 (higher score indicates greater loneliness)
[j] This measure was added to the baseline following study launch (Supplement 1). Missing data are due to timing of administration

SMD: Standardized Mean Difference
P-nom: Nominal P-value
P-adj: Multiplicity-Adjusted P-value (via Holm's procedure)

eTable 5. Vaping Cessation Outcomes Among 7-month Responders, % (95% CI)

Outcome Variable	Intervention (n=759)	Control (n=744)	Rate Difference (95% CI)	Rate Ratio (95% CI)	Odds Ratio (95% CI)	P-val
30-day ppa						
No. responses	521	543				
No. abstinent	287	208				
CCA	55.1 (50.6, 60.0)	38.3 (34.7, 42.3)	16.8 (10.9, 22.7)	1.44 (1.26, 1.64)	1.98 (1.55, 2.52)	<.001
IPRW	53.9 (49.6, 58.2)	37.9 (33.8, 42.0)	16.0 (10.0, 22.0)	1.42 (1.24, 1.63)	1.92 (1.50, 2.24)	<.001
Missing=Vaping	37.8 (34.4, 41.3)	28.0 (24.9, 31.3)	9.9 (5.1, 14.5)	1.35 (1.17, 1.57)	1.57 (1.26, 1.95)	<.001
Repeated ppa						
No. responses	517	538				
No. abstinent	131	61				
CCA	25.3 (21.7, 29.6)	11.3 (9.0, 14.2)	14.0 (9.4, 18.6)	2.24 (1.70, 2.94)	2.65 (1.90, 3.70)	<.001
IPRW	25.1 (21.6, 29.2)	11.4 (9.0, 14.4)	13.7 (9.0, 18.4)	2.21 (1.67, 2.93)	2.61 (1.87, 3.65)	<.001
Missing=Vaping	17.3 (14.7, 20.1)	8.2 (6.4, 10.4)	9.1 (5.7, 12.4)	2.10 (1.58, 2.80)	2.34 (1.69, 3.22)	<.001

ppa: point prevalence abstinence

CCA: Complete Case Analysis

IPRW: inverse probability of retention weighting

eTable 6. Moderators of Intervention Effects on 30-day Point Prevalence Abstinence Rates

	Beta	Std. Error	P-nom	P-adj
Demographic characteristics				
Age	.008	.130	.949	1.000
Grade level (Ref: 6th-8th)			*.292*	1.000
9th	-.185	1.149	.872	
10th	.982	1.024	.337	
11th	.820	1.006	.415	
12th	.747	1.005	.458	
>12th Grade	.647	1.277	.612	
Ungraded or Other Grade/Not a student	-.358	1.168	.759	
Gender (Ref: Male)			*.950*	1.000
Female	-.053	.232	.820	
Non-binary or Other	.067	.437	.878	
LGBQ+ (Ref: Heterosexual)	-.148	.229	.519	1.000
Racial minority (Ref: White)	.514	.227	.024	.552
Hispanic ethnicity (Ref: non-Hispanic)	.187	.306	.542	1.000
Vaping-related items				
Vaping 21+ days per month (Ref: 20 days or less)	-.477	.312	.126	1.000
Motivation to quit vaping	.288	.135	.033	.693
Confidence to quit vaping	.169	.104	.103	1.000
Past year attempt to quit vaping (Ref: None)			*.436*	1.000
1-2 times	-.337	.378	.372	
3-5 times	.030	.366	.935	
6 or more times	-.361	.439	.412	
Concern about health consequences of vaping	.021	.095	.826	1.000
Nicotine/e-cigarette dependence				
10-item Penn State E-Cig Dependence Index	-.017	.027	.519	1.000
E-cigarette Dependence Scale	.034	.033	.298	1.000
E-cigarette Fagerström Test of Cigarette Dependence	-.036	.050	.463	1.000
Hooked on Nicotine Checklist	-.032	.048	.500	1.000
Vape within 30 minutes after waking	-.281	.254	.269	1.000
Perceived addiction to vaping (Ref: Very Addicted)			*.026*	.572
Somewhat addicted	.465	.713	.515	
Not at all addicted	.056	.720	.938	
I don't know	1.653	.896	.065	
Psychosocial characteristics				
Global Assessment of Individual Needs - Short Screener (GAIN-SS)				
Internalizing Disorders	-.313	.121	.010	.250
Substance Disorder Problems	-.027	.067	.685	1.000
Loneliness Scale	-.061	.035	.080	1.000
PEARLS (Ref: Low Risk)			*.482*	1.000
Intermediate risk	.176	.291	.544	
High risk	.626	.572	.274	
Other substance use				
Past 30-day use cigarettes	.346	.237	.144	1.000
Past 30-day use large cigars, little cigars, cigarillos	-.024	.328	.942	1.000
Past 30-day use nicotine pouches	-.362	.348	.298	1.000
Past 30-day use marijuana/cannabis	-.012	.254	.962	1.000

P-nom: Nominal P-value
P-adj: Multiplicity-Adjusted P-value (via Holm's procedure)
PEARLS: PEdiatric Adverse childhood events and Related Life events Screener

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